

Notes: Schoar (JF, 2002)
Effects of Corporate Diversification on Productivity

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Plant-level data and sample construction	4
3.2	Total factor productivity	4
3.3	Diversification measures	5
3.4	Acquisitions, incumbent plants, and new plants	5
3.5	Valuation, overhead, and wage variables	6
4	Methodology and empirical specifications	6
4.1	Static productivity regression	6
4.2	Plant fixed effects	6
4.3	Acquired-plant event study	7
4.4	Incumbent-plant event study	7
4.5	Robustness checks	7
4.6	Market valuation and wage regressions	8
5	Identification and interpretation	8
5.1	What the paper can identify	8
5.2	Static selection versus dynamic effect	8
5.3	Acquisition selection concern	8
5.4	New toy effect versus optimal refocusing	9
5.5	Measurement issues	9
5.6	Interpretation of the diversification discount	9
6	Tables and figures	10
6.1	Table I: Descriptive statistics	10
6.2	Table II: Effect of diversification on productivity	10
6.3	Table III: Effect of ownership changes on acquired plants	11
6.4	Table IV: Effect of diversification on incumbent plants	11

6.5	Table V: Overhead costs	12
6.6	Table VI: Excess value and productivity	13
6.7	Table VII: Excess value and diversification	13
6.8	Table VIII: Wages	14
7	Short summary	14
8	Conclusion	15
8.1	Core conclusion	15
8.2	Economic intuition	15
8.3	Incremental contribution revisited	15
8.4	Implications	15
8.5	Limitations	16
8.6	Takeaway	16

1 Big picture

- **Research question.** What are the real productivity effects of corporate diversification? More specifically, are conglomerates less efficient than stand-alone firms, or does diversification affect productivity through a more subtle channel?
- **Main idea.** The diversification discount in market values could reflect inefficient conglomerates, measurement problems in segment data, selection into diversification, or rent dissipation. Schoar opens the “black box” of conglomerates using plant-level production data rather than relying only on stock market valuations.
- **Empirical object.** The paper measures total factor productivity (TFP) at the plant level using the Longitudinal Research Database (LRD), matched to publicly traded Compustat firms.
- **Static finding.** Plants in diversified firms are, on average, more productive than comparable plants in stand-alone firms. The estimated productivity premium is economically large: moving from low diversification to the sample mean implies roughly a 7 percent productivity increase.
- **Dynamic finding.** When firms become more diversified, productivity falls at the firm level. Newly acquired plants improve, but incumbent plants deteriorate. Because incumbent plants are numerous, the net firm-level effect is negative.
- **Core mechanism.** The paper describes a “new toy” effect: managers appear to shift attention and resources toward newly acquired segments, improving those new plants while neglecting existing plants.
- **Valuation finding.** Stock prices track productivity both in the cross section and within firms over time. The tracking is not weaker for conglomerates, so lower stock market transparency is unlikely to be the main explanation for the diversification discount.
- **Rent dissipation.** Diversified firms do not seem to hide productivity problems through omitted corporate overhead. Instead, they appear to pay workers higher wages and benefits, suggesting that some productivity gains may be captured by employees rather than shareholders.
- **Economic takeaway.** Diversified firms are not necessarily inefficient in levels, but diversification as a corporate strategy can destroy value dynamically by diluting managerial attention and shifting rents away from shareholders.

2 Incremental contribution of the paper

- **From valuation discount to real operations.** Earlier work focuses heavily on the diversification discount in market values. Schoar studies whether diversification is associated with real productivity differences inside the firm.
- **Plant-level evidence.** The paper uses LRD plant-level data, which provide a more detailed view of conglomerates than Compustat business-segment data.
- **Static versus dynamic distinction.** The paper separates the productivity level of diversified firms from the productivity effect of becoming more diversified. This distinction is central: diversified firms look good in the cross section, but diversifying moves reduce productivity.

- **Incumbents versus new plants.** The paper decomposes the effect of diversification into acquired plants and incumbent plants. This makes the managerial-attention channel visible.
- **Market valuation link.** The paper connects plant-level productivity to excess value measures, showing that market valuations do contain information about real productivity.
- **Transparency test.** If conglomerate stock prices were less informative, the link between productivity and market value should be weaker for diversified firms. The paper finds the opposite, which weakens a transparency-based explanation.
- **Rent distribution insight.** By studying wages, the paper shows that shareholder value need not rise one-for-one with productivity. A firm can be more productive and still trade at a discount if rents are dissipated or redistributed.
- **Governance interpretation.** The results point toward agency problems inside diversified firms: managers may overemphasize new divisions, underinvest attention in existing divisions, and allow rents to be shared with employees.

3 Data and measurement

3.1 Plant-level data and sample construction

The main data source is the Longitudinal Research Database (LRD) at the U.S. Bureau of the Census. The LRD contains plant-level information for manufacturing establishments, with SIC codes from 2000 to 3999.

- **Sample period.** The plant-level sample covers 1977 to 1995.
- **Public-firm match.** The paper uses LRD plants belonging to publicly traded firms that can be matched to Compustat.
- **Compustat timing.** The LRD-Compustat match includes firms listed in Compustat in 1987, then links their corresponding plants across sample years.
- **Main advantage.** The data identify individual plant performance, including the performance of acquired plants before and after ownership changes.
- **Main limitation.** The LRD covers manufacturing establishments. Diversification outside manufacturing must be supplemented with Compustat segment information.
- **Scale.** Table I reports 245,006 plant-level observations and 20,366 firm-level observations.

Interpretation. The data are unusually powerful because they let the paper study what happens inside conglomerates, not just how investors value conglomerates from the outside.

3.2 Total factor productivity

The primary performance measure is plant-level total factor productivity at the three-digit SIC industry level. For each industry and year, the paper estimates a Cobb-Douglas production function:

$$\ln(y_{ijt}) = a_{jt} + b_{jt} \ln(K_{ijt}) + c_{jt} \ln(L_{ijt}) + d_{jt} \ln(M_{ijt}) + \varepsilon_{ijt},$$

where y_{ijt} is plant output, K_{ijt} is capital, L_{ijt} is labor, and M_{ijt} is materials.

- **Output.** Output is plant sales, measured as total value of shipments, plus changes in finished-goods and work-in-process inventories.
- **Labor.** Labor is production-worker equivalent man-hours.
- **Capital.** Capital stock is constructed using a perpetual-inventory method starting from the earliest available book value.
- **Materials.** Materials include parts, intermediate goods, fuel, energy, and contracted-work inputs.
- **TFP.** The residual ε_{ijt} is the plant's TFP. It measures the plant's productivity relative to other plants in the same three-digit industry and year.

Interpretation. Because the production-function coefficients vary by industry and year, TFP is meant to compare plants within a narrow technology group rather than across fundamentally different industries.

3.3 Diversification measures

Diversification is measured using firm identifiers in the LRD and supplemented with Compustat segment data for nonmanufacturing segments.

- **Two-digit diversification.** The main measure is the logarithm of the number of two-digit SIC segments in which the firm operates.
- **Herfindahl diversification.** The paper also uses one minus a Herfindahl index based on segment weights, where weights are based on shipments or capital stock.
- **Stand-alone firms.** Firms operating in only one two-digit industry are classified as stand-alone.
- **Diversified firms.** Multisegment firms are split into diversified and highly diversified firms depending on whether their number of segments is below or above the diversified-firm mean.

Interpretation. The two-digit SIC measure is simple and transparent. The Herfindahl measure additionally captures whether activity is concentrated in one segment or spread across several segments.

3.4 Acquisitions, incumbent plants, and new plants

The dynamic analysis uses plant ownership changes and diversifying events.

- **Acquired plants.** Plants that change owners between 1977 and 1995 are used to study whether new owners improve plant productivity.
- **Diversifying acquisitions.** A diversifying acquisition is an acquisition of a plant in a two-digit industry in which the acquirer did not previously operate.
- **Related acquisitions.** Related acquisitions are used as a comparison group for acquisitions that do not change the firm's diversification status.
- **Event window.** The paper typically uses a three-year window before and after the ownership change or diversifying event.

- **Incumbent plants.** Incumbent plants are plants owned by the firm before the diversifying move. These are the plants most informative about whether diversification distracts managers from existing divisions.

3.5 Valuation, overhead, and wage variables

The paper combines plant-level productivity with financial and compensation measures.

- **Excess value.** Excess value is the logarithm of a firm’s Tobin’s q relative to an imputed benchmark based on size-weighted industry qs .
- **Corporate overhead.** The paper uses the Central Auxiliary Organizations (CAO) survey for 1977, 1982, 1987, and 1992 to measure auxiliary facilities such as headquarters and off-site administrative units.
- **Wages.** Hourly wages are constructed from the production-worker wage bill divided by production-worker hours. A broader wage measure adds legally required and voluntary supplementary labor costs.

Interpretation. These additional variables help distinguish several explanations for the diversification discount: hidden overhead, misvaluation, and redistribution of rents to workers.

4 Methodology and empirical specifications

4.1 Static productivity regression

The baseline plant-level productivity regression is:

$$TFP_{ijt} = a + bDIV_{it} + cSIZE_{it} + dAGE_{it} + \varepsilon_{ijt}.$$

Here DIV_{it} is a diversification measure, $SIZE_{it}$ includes firm and segment size controls, and AGE_{it} is plant age.

- The unit of observation is the plant-year.
- TFP is winsorized at the 1st and 99th percentiles.
- Segment size and firm size are measured by number of plants and entered as splines.
- Standard errors are corrected for clustering at the firm-year level.

Interpretation. This regression asks whether plants belonging to diversified firms are more productive than otherwise similar plants.

4.2 Plant fixed effects

The paper then adds plant fixed effects. This changes the interpretation from a cross-sectional comparison to a within-plant comparison.

- Without plant fixed effects, diversification is positively related to productivity.

- With plant fixed effects, the coefficient on diversification turns negative.
- This means the static premium is driven by heterogeneity across plants, while increases in diversification are associated with productivity declines within plants.

Interpretation. This is the paper’s first major distinction between being diversified and becoming diversified.

4.3 Acquired-plant event study

For plants that change owners, the paper estimates:

$$TFP_{ijt} = \alpha_i + bAFTER + c(AFTER \times DIVERS) + dDIVERS + eSIZE_{it} + \varepsilon_{ijt}.$$

- *AFTER* equals one in the three years after the acquisition.
- *DIVERS* equals one when the plant moves from a less diversified to a more diversified owner.
- Plant fixed effects absorb time-invariant plant quality.
- The specification asks whether acquired plants become more productive after changing owners, especially when the new owner is more diversified.

Interpretation. This test addresses the concern that diversified firms simply buy already productive plants. If that were the whole story, acquired plants should not improve after acquisition.

4.4 Incumbent-plant event study

For incumbent plants of diversifying firms, the paper estimates a similar before-after regression and compares diversifying acquisitions to related acquisitions:

$$TFP_{ijt} = \alpha_i + bAFTER + c(AFTER \times DIVERS) + dDIVERS + eSIZE_{it} + \varepsilon_{ijt}.$$

- The sample includes plants already owned before the diversifying move.
- Related acquisitions provide a comparison group for the effect of any acquisition.
- Diversifying events must be economically meaningful: the new addition must be at least 10 percent of the firm’s pre-existing capital stock.
- Plants divested within three years after the acquisition are excluded, which should bias against finding a decline because divested plants tend to have deteriorating productivity.

Interpretation. This specification isolates whether existing plants suffer after the firm expands into a new industry.

4.5 Robustness checks

The paper studies two plant-level measurement concerns.

- **Omitted overhead.** If diversified firms move overhead out of manufacturing plants, plant-level TFP could be overstated. The CAO data show that diversified firms do not use more excess overhead; if anything, they use less.

- **Transfer pricing.** If internal transfers are priced strategically, downstream or upstream plants could appear artificially productive. The paper uses input-output data to identify vertical relatedness and finds no evidence that transfer pricing explains the productivity premium.

4.6 Market valuation and wage regressions

The paper then links real productivity to shareholder value and rent distribution.

- **Excess value and productivity.** Firm-level TFP is constructed as a weighted average of plant-level TFP. Excess value is then regressed on firm TFP.
- **Diversification discount.** Excess value is regressed on diversification measures computed from both LRD and Compustat segment definitions.
- **Wages.** Production-worker wages are regressed on diversification and controls for segment size, firm size, plant age, capital, industry fixed effects, and year fixed effects.

Interpretation. The valuation analysis asks whether markets recognize productivity. The wage analysis asks whether productivity gains accrue partly to workers rather than shareholders.

5 Identification and interpretation

5.1 What the paper can identify

The paper provides strong descriptive and event-based evidence about how plant-level productivity differs by diversification status and how productivity changes around diversifying moves. The evidence is not based on randomized diversification, so the interpretation remains partly observational. However, plant fixed effects, event windows, related-acquisition comparisons, and robustness checks substantially sharpen the interpretation.

5.2 Static selection versus dynamic effect

The static productivity premium does not mean that diversification causes plants to become more productive. It means that plants inside diversified firms are more productive in the cross section. The plant-fixed-effect and event-study results then show that, when firms increase diversification, existing productivity tends to fall.

Key distinction.

- **Level effect:** diversified firms own productive plants.
- **Change effect:** becoming more diversified reduces productivity, especially among incumbent plants.

5.3 Acquisition selection concern

A natural concern is that diversified firms buy high-productivity plants and then later run them down. The acquired-plant results do not support this explanation.

- Acquired plants have low and declining productivity before the ownership change.
- Plants bought by diversified firms do not have higher ex ante productivity than plants bought by stand-alone firms.
- Acquired plants improve after acquisition, especially in diversifying acquisitions.

Interpretation. Diversified firms appear capable of improving newly acquired assets. The problem arises from what happens to existing assets.

5.4 New toy effect versus optimal refocusing

The “new toy” interpretation is that management attention shifts toward new segments. An alternative is that managers diversify rationally because existing segments are deteriorating.

The paper argues against the optimal-refocusing interpretation:

- The industry Q of new segments is below the average industry Q of the firm’s initial segments.
- Incumbent plants are improving before diversification and decline afterward.
- Market values decline after diversifying moves rather than improving.

Interpretation. The evidence is more consistent with agency or attention problems than with efficient reallocation away from poor opportunities.

5.5 Measurement issues

Several measurement concerns remain important.

- TFP residuals can reflect markups as well as physical efficiency if product markets are not perfectly competitive.
- The LRD sample is limited to manufacturing plants and firms present in the 1987 Compustat match.
- Diversification outside manufacturing is imperfectly measured, although the paper supplements with Compustat segment data.
- Plant-level data may omit headquarters or auxiliary activities, but the CAO analysis reduces this concern.
- Internal transfer pricing may distort plant-level productivity, but the vertical-relatedness tests reduce this concern.

5.6 Interpretation of the diversification discount

The paper does not argue that the diversification discount is mechanically caused by low productivity. Instead, diversified firms have higher productivity in levels. The discount can coexist with a productivity premium because:

- diversifying moves reduce productivity dynamically;

- stock prices may price expected future inefficiencies;
- productivity gains may be shared with workers through higher wages and benefits;
- agency problems may distort how surplus is allocated inside the firm.

Interpretation. Higher productive efficiency does not automatically imply higher shareholder value.

6 Tables and figures

6.1 Table I: Descriptive statistics

Read:

- Table I compares all firms, stand-alone firms, diversified firms, and highly diversified firms.
- Diversified firms operate many more plants than stand-alone firms: stand-alone firms average about 3.5 plants, diversified firms about 10.3 plants, and highly diversified firms about 42.6 plants.
- Plant size differs much less than firm size. The average plant in a diversified firm is only moderately larger than the average plant in a stand-alone firm.
- Diversified firms are more capital intensive than stand-alone firms and pay higher production wages.
- The table also shows that output per worker is higher in diversified firms, while output per capital is slightly lower.

Economic message: Diversified firms are much larger organizations, but the average plant-level comparison is not simply a comparison of enormous plants to tiny plants. The table motivates the need for TFP rather than partial productivity measures.

6.2 Table II: Effect of diversification on productivity

Read:

- Table II is the central static productivity table.
- Without plant fixed effects, diversification is positively related to plant TFP.
- The two-digit diversification coefficient is positive and significant in the baseline specifications.
- The Herfindahl diversification measure gives the same qualitative conclusion.
- When plant fixed effects are included, the diversification coefficient becomes negative.
- Controlling for vertical relatedness with upstream and downstream dummies does not remove the positive diversification coefficient.

Economic message: Diversified firms own more productive plants, but increases in diversification are associated with declines within plants. This table introduces the paper’s key static-versus-dynamic contrast.

Table I
Descriptive Statistics

Plant-level characteristics for the sample firms in the LRD between 1977 and 1995. Total value of shipments and capital stock are in million dollars, hours worked in millions, and production wages are in dollars. Total value of shipments is plant sales in the LRD and contains interfirm transfers valued at market prices. Capital stock includes book values of machinery and buildings. Hours worked is defined as production-worker hours per year. Diversification is defined as number of segments at the two-digit SIC level. Stand-alone firms operates in only one industry. Multisegment firms are subdivided into diversified firm (have more than one but less than the mean number of segments of diversified firms) versus highly diversified firms (above mean number of segments).

	Sample							
	All Firms		Stand-Alone		Diversified		Highly Diversified	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Plant level:								
Total value of shipments	40.43	43.91	38.53	41.56	41.25	44.30	40.80	43.46
Capital stock	16.70	14.78	13.03	14.65	17.77	14.79	16.81	14.86
Hours worked	0.26	0.25	0.25	0.24	0.25	0.25	0.27	0.26
Capital/worker	55.08	46.47	51.59	46.65	57.16	47.07	53.71	45.61
Production wage	10.45	3.69	10.08	3.72	10.62	3.71	10.35	3.66
Total value of shipments/hour	172.59	123.56	163.31	122.07	182.59	128.57	164.95	118.09
Total value of shipments/capital	7.82	5.65	8.39	5.69	7.87	5.66	7.61	5.62
Number of observations	245,006		23,770		112,669		108,567	
Firm Level:								
Plants per firm	12.03	20.38	3.46	5.76	10.29	14.21	42.64	34.89
Plants per segment	6.31	10.19	3.46	5.76	6.28	9.98	14.17	15.09
Two-digit diversification	2.87	2.30	1.00	0.00	2.90	1.97	7.82	1.98
Number of observations	20,366		6,869		10,951		2,546	

Table I: Descriptive statistics from the LRD plant and firm sample.

Table II
Effect of Diversification on Productivity

The dependent variable is total factor productivity (TFP) at the three-digit SIC level. TFP is the residual from estimating a log linear Cobb–Douglas production function for each industry and year at the plant level, where one regresses the value of output (total value of shipments adjusted for changes in inventories) on labor (production worker hours), capital stock (constructed via the perpetual inventory method), and material inputs (intermediate inputs, fuels, and energy consumed). Two-digit diversification is the logarithm of the number of segments at the two-digit SIC level. Herfindahl is an assets-weighted measure of diversification. The dummy variable Down is equal to one if a plant is a downstream segment in a vertically integrated firm. Similarly, Up equals one if a plant is an upstream segment in a diversified firm, and zero otherwise. Controls for Segment size and Firm size are constructed as the number of plants at the segment or firm level, respectively. These size controls are splined. Age is the logarithm of plant age. Heteroskedasticity-robust standard errors are in parentheses. Standard errors are corrected to allow for group effects within firms.

	(1)	(2)	(3)	(4)	(5)	(6)
Two-digit diversification	0.064*** (0.015)		0.034** (0.016)		−0.070** (0.030)	0.042** (0.020)
Herfindahl		0.154*** (0.043)		0.134** (0.049)		
Age	0.142*** (0.012)	0.142*** (0.012)	0.135*** (0.012)	0.136*** (0.012)	−0.018*** (0.010)	0.135*** (0.012)
Segment size1			0.167*** (0.029)	0.165*** (0.028)	−0.047 (0.035)	0.168*** (0.029)
Segment size2			0.158*** (0.036)	0.161*** (0.036)	−0.082 (0.039)	0.157*** (0.036)
Segment size3			0.080 (0.059)	0.087 (0.059)	0.113* (0.058)	0.081 (0.058)
Segment size4			−0.79 (0.070)	−0.066 (0.070)	−0.017 (0.067)	−0.082 (0.070)
Segment size5			−0.034 (0.072)	−0.017 (0.072)	0.089 (0.083)	−0.036 (0.072)
Firm size1			0.014 (0.026)	0.010 (0.024)	0.080** (0.034)	0.010*** (0.009)
Firm size2			0.016 (0.050)	0.016 (0.050)	0.136** (0.051)	0.008 (0.050)
Firm size3			−0.007 (0.078)	−0.008 (0.076)	−0.136* (0.067)	−0.013 (0.078)
Firm size4			−0.009 (0.101)	−0.024 (0.101)	0.191** (0.083)	−0.011 (0.100)
Firm size5			−0.037 (0.109)	−0.042 (0.108)	0.112 (0.089)	−0.045 (0.110)
Down						0.025 (0.035)
Up						0.019 (0.042)
Plant fixed effects	No	No	No	No	Yes	No
Adjusted R^2	0.035	0.036	0.041	0.042	0.542	0.561
Number of observations	245,006	245,006	245,006	245,006	245,006	245,006

The symbols ***, **, and * indicate statistical significance at the 1, 5, and 10 percent levels, respectively.

Table II: Plant-level TFP regressions of productivity on diversification.

6.3 Table III: Effect of ownership changes on acquired plants

Read:

- Table III studies plants that change owners in a three-year window around acquisition.
- The coefficient on *AFTER* is positive for all acquisitions, suggesting that acquired plants improve after ownership changes.
- The interaction *DIVERS* $\times$ *AFTER* is positive and significant, meaning plants moving to more diversified owners improve more.
- For the subsample of diversifying acquisitions, acquired plants show a roughly 3 percent productivity increase after the acquisition.

Economic message: Diversified firms do not merely buy good plants and run them down. Newly acquired plants tend to improve after being acquired, especially in diversifying deals.

6.4 Table IV: Effect of diversification on incumbent plants

Read:

- Table IV studies plants already owned by the firm before a diversifying move.

- Diversifying events have a more negative effect on incumbent plants than related acquisitions.
- In the diversifying-acquisition subsample, incumbent plant productivity declines by about 2.6 percent after diversification.
- At the firm level, total firm TFP declines by about 2.3 percent after diversification.

Economic message: The new plants improve, but the old plants suffer. This is the core evidence for the “new toy” effect and for dynamic inefficiency from diversification.

Table III
Effect of Ownership Changes on Acquired Plants

The sample contains only observations for plants that change owners in the three periods before and after the acquisition. Columns (1) and (2) include diversifying as well as related acquisition events, column (3) is based on the subsample of diversifying acquisitions. A diversifying move is characterized as an acquisition of a plant in a two-digit industry other than the existing industries of the firm. Only acquisitions that constitute at least 10 percent of the firm's pre-existing capital stock are included. The dependent variable is total factor productivity (TFP) at the three-digit SIC level (as described in Table II). After is a dummy variable equal to one in the three periods after the acquisition and zero in the three periods before. Divers is a dummy variable equal to one, if the acquired plant moves from a less to a more diversified firm, and zero otherwise. Controls for Segment size and Firm size are constructed as the number of plants at the segment level or the firms level, respectively. The size variables are splined. Heteroskedasticity-robust standard errors are in parentheses. Standard errors are corrected to allow for group effects within firms, since diversification varies only at the firm level.

	Specification		
	All Acquisitions		Only Diversifications
	(1)	(2)	(3)
After	0.004*** (0.000)	0.000 (0.004)	0.029** (0.013)
Divers * After		0.006*** (0.002)	
Segment size1	0.031* (0.016)	0.033** (0.016)	0.033 (0.027)
Segment size2	0.010 (0.021)	-0.009 (0.021)	-0.087* (0.046)
Segment size3	-0.006 (0.029)	-0.003 (0.037)	0.033* (0.019)
Segment size4	-0.007 (0.028)	-0.004 (0.029)	-0.071*** (0.017)
Segment size5	0.040 (0.042)	0.048 (0.042)	0.045 (0.050)
Firm size1	-0.011 (0.013)	-0.012 (0.013)	-0.060 (0.039)
Firm size2	0.011 (0.017)	0.012 (0.018)	0.052 (0.039)
Firm size3	-0.006 (0.010)	-0.009 (0.010)	-0.013 (0.025)
Firm size4	-0.014 (0.011)	-0.016 (0.011)	-0.039 (0.030)
Firm size5	0.013 (0.010)	0.010 (0.010)	0.024 (0.023)

Table III: Acquired plants improve after ownership changes, especially in diversifying acquisitions.

Table IV
Effect of Diversification on TFP of Incumbent Plants

The dependent variable is total factor productivity (TFP) at the three-digit SIC level (as described in Table II). After is a dummy variable equal to one in the three periods after a diversifying move and equal to zero in the three periods before. Controls for Segment size and Firm size are constructed as the number of plants at the segment level or the firm level, respectively. The size variables are splined. The sample in columns (1) and (2) contains observations for the incumbent plants of a firm in the three periods before and after a related or diversifying acquisition. The results in column (3) are based only on the subsample of diversifying acquisitions. The sample in column (4) contains *firm* level observations, where firm TFP indexes are formed as the weighted average of the plant level TFP measures. A diversifying event is defined as an acquisition of a plant in a two-digit industry other than the firm's existing industries. Heteroskedasticity-robust standard errors are in parentheses. Standard errors are corrected to allow for group effects within firms.

	Specification			
	All Acquisitions		Only Diversifications	
	(1)	(2)	(3)	(4)
After	-0.002 (0.002)	0.001 (0.003)	-0.026*** (0.004)	-0.023*** (0.006)
Divers		0.002 (0.005)		
Divers * After		-0.010** (0.005)		
Segment size1	0.025** (0.012)	0.024* (0.012)	0.032* (0.018)	
Segment size2	-0.001 (0.016)	-0.001 (0.017)	-0.026 (0.024)	
Segment size3	-0.010 (0.007)	-0.010 (0.007)	0.001 (0.009)	
Segment size4	0.007 (0.008)	0.006 (0.009)	0.016 (0.012)	
Segment size5	0.017* (0.009)	0.016* (0.009)	0.013 (0.013)	
Firm size1	-0.003 (0.011)	-0.003 (0.011)	-0.006 (0.039)	0.009 (0.015)
Firm size2	0.006 (0.016)	0.006 (0.016)	0.052 (0.039)	-0.052** (0.019)
Firm size3	0.003 (0.008)	0.004 (0.008)	-0.013 (0.025)	0.002 (0.013)
Firm size4	-0.008 (0.009)	-0.006 (0.009)	-0.039 (0.030)	-0.006 (0.016)
Firm size5	-0.003 (0.011)	-0.002 (0.011)	0.024 (0.023)	0.007 (0.017)
Plant fixed effects	Yes	Yes	Yes	No
Firm fixed effects	No	No	No	Yes
Adjusted R ²	0.537	0.536	0.572	0.545
Number of observations	51,313	51,313	23,065	2,855

The symbols ***, **, and * indicate statistical significance at the 1, 5, and 10 percent level, respectively.

Table IV: Incumbent plants lose productivity after diversifying moves.

6.5 Table V: Overhead costs

Read:

- Table V uses CAO data on auxiliary facilities to test whether diversified firms appear productive only because overhead costs are outside manufacturing plants.
- The dependent variables are excess overhead inputs for labor, capital, and materials.
- The diversification coefficients are negative across labor, capital, and material overhead measures.
- The results remain negative after controlling for firm size.

Economic message: Omitted headquarters or auxiliary inputs do not explain the productivity

premium of diversified firms. If anything, diversified firms use less measured overhead relative to benchmarks.

6.6 Table VI: Excess value and productivity

Read:

- Table VI relates excess value to aggregate firm TFP.
- Firm TFP is positively and significantly related to excess value in the cross section.
- With firm fixed effects, changes in firm TFP are also positively related to changes in excess value.
- Interactions between diversification and TFP are positive, not negative.
- The direct diversification effect remains negative, indicating a discount even after accounting for productivity.

Economic message: Stock prices do reflect productivity, and the reflection is not weaker for diversified firms. The diversification discount is therefore hard to explain solely by lower transparency.

Table V
Overhead Costs

Data on overhead inputs are obtained from the CAO (Central Auxiliary Organizations) database at the Bureau of the Census for the years 1977, 1982, 1987, and 1992. Separate information on labor, capital, and material used are available. The dependent variables, Excess overhead costs, are measured as the difference between the ratio of productive inputs in auxiliary facilities and the LRD and compared to a benchmark of imputed inputs. The benchmark of imputed inputs is constructed as the weighted sum of inputs from the median stand-alone firm in the corresponding industry. Two-digit diversification is the logarithm of the number of segments at the two-digit SIC level. Firm size is measured as the total value of capital at the firm level. Heteroskedasticity-robust standard errors are in parentheses.

	Specification					
	Labor		Capital		Material	
	(1)	(2)	(3)	(4)	(5)	(6)
Two-digit diversification	-0.018*** (0.003)	-0.016*** (0.004)	-0.003** (0.001)	-0.003* (0.002)	-0.001* (0.000)	-0.001* (0.000)
Firm size		-0.002 (0.002)		-0.001 (0.001)		-0.001** (0.000)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.073	0.074	0.069	0.069	0.089	0.089
Number of observations	3,625	3,625	3,625	3,625	3,625	3,625

The symbols ***, **, and * indicate statistical significance at the 1, 5, and 10 percent levels.

Table V: Diversified firms do not have higher excess overhead costs.

Table VI
Excess Value and Productivity

The dependent variable is excess value (the logarithm of the ratio of a firm's individual Tobin's q to its imputed benchmark of size weighted industry qs from COMPUSTAT industry segment files). Two-digit diversification is the logarithm of the number of segments at the two-digit level. Herfindahl is an assets-weighted measure of diversification. Firm TFP is the weighted average of plant level total factor productivity (TFP) at the two-digit SIC level. Controls for Firm size are constructed as the number of plants at the firm level. Heteroskedasticity-robust standard errors are in parentheses.

	(1)	(2)	(3)	(4)	(5)	(6)
Firm TFP	0.374*** (0.021)	0.365*** (0.021)	0.106*** (0.021)	0.108*** (0.021)	0.276*** (0.028)	0.280*** (0.024)
Two-digit diversification					-0.043*** (0.009)	
Two-digit div * TFP					0.178*** (0.038)	
Herfindahl						-0.100*** (0.022)
Herfindahl * TFP						0.756*** (0.110)
Firm size		0.012** (0.004)		0.016** (0.006)	0.024*** (0.005)	0.018*** (0.004)
Firm fixed effect	No	No	Yes	Yes	No	No
Adjusted R^2	0.053	0.054	0.567	0.567	0.061	0.065
Number of observations	8,561	8,561	8,561	8,561	8,561	8,561

The symbols ***, **, and * indicate statistical significance at the 1, 5, and 10 percent level, respectively.

Table VI: Firm TFP is positively related to excess value.

6.7 Table VII: Excess value and diversification

Read:

- Table VII recomputes the diversification discount using both LRD and Compustat segment measures.
- Diversification is negatively related to excess value using LRD-based measures.
- The discount is also present using Compustat-based measures.
- The LRD-based discount is somewhat smaller, but still economically meaningful.

Economic message: Segment-data measurement problems may affect magnitudes, but they do not eliminate the diversification discount in this sample.

6.8 Table VIII: Wages

Read:

- Table VIII studies whether diversified firms pay higher production-worker wages.
- The coefficient on diversification is positive and significant in all specifications.
- The wage premium survives controls for segment size, firm size, plant age, capital, and output per worker.
- When supplementary labor costs are included, the diversification coefficient is even larger.
- The paper interprets the estimates as implying roughly 8 percent higher wages in diversified firms for a one-standard-deviation increase in diversification.

Economic message: Some of the productivity advantage of diversified firms may be shared with workers rather than captured by shareholders. This can help explain why productivity and shareholder value move differently.

Table VII
Excess Value and Diversification

The dependent variable is excess value (calculated as the logarithm of the ratio of a firm's individual Tobin's q to its imputed benchmark of size weighted industry q s). Two different sources of segment measures are used: Excess Value1 is based on segment measures from the LRD and Excess Value2 uses information from COMPUSTAT industry segment files. Two-digit diversification is the logarithm of the number of segments at the two-digit level. Herfindahl is an assets-weighted measure of diversification. Segnum is the number of segments at the two-digit SIC level according to COMPUSTAT. Controls for Firm size are constructed as the number of plants at the firm level. Heteroskedasticity-robust standard errors are in parentheses.

	Dependent Variable					
	Excess Value1 (LRD)			Excess Value2 (COMPUSTAT)		
	(1)	(2)	(3)	(4)	(5)	(6)
Two-digit diversification	-0.045*** (0.009)			-0.072*** (0.018)		
Herfindahl		-0.081*** (0.022)			-0.141*** (0.044)	
Segnum			-0.095*** (0.009)			-0.147*** (0.018)
Firm size	0.036*** (0.005)	0.029*** (0.004)	0.039*** (0.004)	0.009 (0.010)	0.003 (0.009)	0.012 (0.008)
Adjusted R^2	0.110	0.101	0.110	0.069	0.068	0.068
Number of observations	8,561	8,561	8,561	8,561	8,561	8,561

The symbols ***, **, and * indicate statistical significance at the 1, 5, and 10 percent levels, respectively.

Table VII: Diversification is negatively related to excess value under both LRD and Compustat measures.

7 Short summary

- The paper studies the real productivity effects of corporate diversification using plant-level Census data.
- Diversified firms are more productive than stand-alone firms in the cross section.

Table VIII
Wages

Dependent variable: production worker wages. Hourly Wage is constructed at the plant level as total wage bill for production worker divided by hours worked in production. Hourly Wage2 includes wages plus additional labor costs measured as the sum of legally required and voluntary supplementary labor costs. Two-digit diversification is the number of segments at the two-digit industry level. Firm size and Segment size are measured as the number of plants per firm or segment, respectively. Age is measured as years since a plant was set up. If the setup date precedes the first year of the sample (1977), that year is taken as the setup year. Total value of shipments per hour worked is calculated at the plant level. Heteroskedasticity-robust standard errors are in parentheses. Standard errors are corrected to allow for group effects within firms, since diversification varies only at firm level.

	Specification			
	Hourly Wage		Hourly Wage2	
	(1)	(2)	(3)	(4)
Two-digit diversification	0.023*** (0.005)	0.022*** (0.005)	0.041*** (0.006)	0.039*** (0.006)
Segment size	0.016*** (0.002)	0.015*** (0.002)	0.014*** (0.002)	0.013*** (0.002)
Firm size	0.011*** (0.003)	0.012*** (0.003)	0.002 (0.003)	0.004 (0.003)
Plant age	0.063*** (0.002)	0.057*** (0.002)	0.054*** (0.003)	0.050*** (0.003)
Capital	0.084*** (0.001)	0.085*** (0.001)	0.093*** (0.001)	0.093*** (0.001)
Total value of shipment/hours worked		0.179*** (0.004)		0.160*** (0.005)
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.539	0.544	0.508	0.509
Number of observations	245,006	245,006	245,006	245,006

The symbols ***, **, and * indicate statistical significance at the 1, 5, and 10 percent levels, respectively.

Table VIII: Diversified firms pay higher wages and broader labor compensation.

- This static productivity premium is not the same as the effect of becoming diversified.
- When a firm diversifies, its incumbent plants experience a productivity decline.
- Newly acquired plants improve after acquisition, especially in diversifying acquisitions.
- The combination of improving new plants and declining old plants is described as a “new toy” effect.
- The net firm-level productivity effect of a diversifying move is negative because incumbent plants dominate the firm.
- Overhead-cost and transfer-pricing checks do not explain the productivity premium.
- Stock prices track firm productivity, including for diversified firms, so lower transparency does not explain the discount on average.
- Diversified firms pay higher wages and benefits, suggesting that some rents may be dissipated or redistributed to workers.
- The main lesson is that diversified firms are not bad in levels, but diversification as a corporate strategy can be bad dynamically.

8 Conclusion

8.1 Core conclusion

Schoar’s central conclusion is that diversified firms are not intrinsically less productive. In the cross section, their plants are more productive than plants in stand-alone firms. However, diversifying moves reduce firm productivity because incumbent plants suffer after the firm expands into new segments.

8.2 Economic intuition

Diversification changes managerial attention. A new segment receives focus, resources, and perhaps talented managers. The acquired plant improves, but the existing plants receive less attention and lose productivity. Since incumbent plants are the bulk of the firm, the net effect is negative.

8.3 Incremental contribution revisited

The paper’s contribution is to separate three ideas that are often mixed together: whether diversified firms are productive, whether diversification improves productivity, and whether shareholders capture the productivity surplus. The answers are different: diversified firms are productive; diversification reduces productivity dynamically; and shareholders may not capture all of the surplus.

8.4 Implications

The results imply that the diversification discount should not be interpreted simply as evidence that conglomerates operate inefficient plants. Instead, conglomerates may own productive assets

but still destroy shareholder value through dynamic attention costs, internal agency problems, and rent sharing with employees.

8.5 Limitations

The paper is observational and focuses on manufacturing plants matched to public firms. TFP residuals can reflect markups as well as technical efficiency. The wage evidence is suggestive because the LRD does not contain detailed worker-level controls such as education or tenure. The paper also cannot fully randomize diversification decisions.

8.6 Takeaway

The enduring takeaway is: diversification is not bad because diversified firms are unproductive; it is bad because diversifying can make firms worse over time and can redirect value away from shareholders. Productivity, valuation, and rent distribution are related but distinct outcomes.